Writing olympiad solutions

L4-ST1 Level 4 Olympiad · Strategy

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How BMO Round 1 is marked and how to write solutions that earn full credit: structure, complete proofs, case analysis, both directions of a "find all" answer, and the classic ways to lose marks.
By the end you can explain how BMO1 scripts are marked and what earns credit; write "find all" solutions that prove both directions with complete case analysis; find and repair the gaps in a flawed proof.
Prerequisites: Integer equations (L3-NT2), Exam craft for the SMC and AMC 12 (L3-ST1). **Contests:** BMO Round 1, Mathematical Olympiad for Girls, USAJMO, BMO Round 2. **Time:** about 75 minutes.

1 Lesson

Consider the following question and two answers to it.

Find all primes p such that p^2+2 is also prime.

Script A. "If $p=2$ we get 6, not prime. If $p=3$ we get 11, prime. If $p=5$ we get 27, and if $p=7$ we get $51=3\times 17$. So the answer is $p=3$."

Script B. "The answer is $p=3$. Indeed $3^2+2=11$ is prime. Conversely, suppose $p\neq 3$. Then $3\nmid p$, so $p\equiv\pm 1 \pmod 3$ and $p^2\equiv 1 \pmod 3$. Hence $3\mid p^2+2$, and since $p^2+2>3$ it is not prime. So $p=3$ is the only such prime."

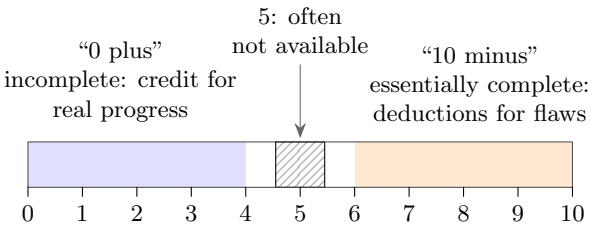
Both have the right answer. On a BMO paper, Script A scores 0 or 1 out of 10 and Script B scores 10. Script A has found the answer; Script B has proved it. This unit is about writing solutions of the second kind.

1.1 How BMO Round 1 is marked

From UKMT's BMO1 pages and the paper's instructions: six problems, three and a half hours, 10 marks each, 60 in total. You must give full written solutions, not just answers, with complete proofs of any assertions; marks depend on the clarity of your presentation, and one complete solution gains more credit than several unfinished attempts.

UKMT's markers' reports explain what happens in practice. Marking is holistic. A script is first judged on whether it is essentially a complete solution. If it is, it is marked down from 10 ("10 minus") for slips, gaps, omitted cases or technical faults. If it is not, it is marked up from 0 ("0 plus"), and partial progress such as particular cases or a significant idea earns at most about 4 marks. On many questions a 5 is not available. The reports also say that correct arguments are rewarded more generously than correct calculations, and that you should aim to finish whole questions rather than collect disconnected parts.

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Compare this with typical scores. On the November 2025 paper the mean was 12.6 out of 60, and the Distinction threshold was 18. Two complete solutions put you in the Distinction band; six half-solutions, each marked "0 plus", might not. Hence the priority is to finish questions.

1.2 The shape of a good solution

1. **State the answer first** in any "find" or "determine" question, in one line.
2. **Say what you will prove**: "We show that n works if and only if $n \equiv 0$ or $3 \pmod 4$." Introduce notation before you use it.
3. **Break the proof into claims**, prove each one and say how they combine.
4. **End with a conclusion** that answers the question as asked.

Rough work stays on rough paper. In geometry, draw a large labelled diagram and make the order of deductions clear: a diagram covered in marked angles with no explanation earns nothing.

1.3 What counts as a complete proof

Every step must follow from earlier steps or from a standard result that you name. "Clearly" is fine only if a reader could fill the gap in five seconds. Reread your solution as a hostile marker, asking "why?" at every line; if the answer is not on the page and is not obvious, put it there. Three failures come up year after year.

Dividing by something that might be zero. Cancelling $x - y$ assumes $x \neq y$. Split into cases instead: either $x = y$, or you may divide.

"Eventually" claims. "Exponentials beat polynomials, so $2^n > n^3$ for large n" is not a proof. Say from which n it holds and prove it, usually by induction.

Careless coprimality. If $ab = 2^k m$ with m odd and $\gcd(a, b) = 1$, then 2^k divides one of a, b. It does not follow that $\{a, b\} = \{2^k, m\}$, because the prime factors of m can be split between a and b.

Example 1.
Find all integers n such that $n^2 + 2$ divides $n^4 + 7$.

Solution.
Small values first: $n = 0, \pm 1, \pm 2$ fail, and $n = \pm 3$ gives $11 \mid 88$, which works. To see whether that is all, remove the leading term by division: $n^4 + 7 = (n^2 + 2)(n^2 - 2) + 11$. The rest follows from this identity.
A good write-up: "The answer is $n = \pm 3$. Since $n^4 + 7 = (n^2 + 2)(n^2 - 2) + 11$, the number $n^2 + 2$ divides $n^4 + 7$ if and only if it divides 11. As $n^2 + 2$ is a positive integer that is at least 2, this happens if and only if $n^2 + 2 = 11$, that is, $n = \pm 3$."
Because each step is an "if and only if", both directions are proved at once; one-way impli-

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cations would only show that ± 3 are the only candidates. And the question says integers, so a script giving only $n = 3$ has lost a solution.

1.4 Case analysis

Cases must cover every possibility, and you must say why they do. Announce the split before you start ("either $p = 2$ or p is odd") and treat each case under its own heading. Small cases matter: if your general argument needs $n \geq 4$, then $n = 1, 2, 3$ must be checked with the working shown. One recent report notes that candidates merely stating they had checked the cases was not enough. You may assume an order such as $a \leq b \leq c$ only when the problem is symmetric, and you must then restore every ordering in the final answer.

Example 2.
Find all triples of positive integers (a, b, c) with $abc = 2(a + b + c)$.

Solution.
The equation is symmetric, so we may assume $a \leq b \leq c$ and permute at the end. The first step is a bound. The right side is at most $6c$, so $abc \leq 6c$ and $ab \leq 6$. That leaves finitely many cases. Rewrite the equation as $c(ab - 2) = 2(a + b)$.
The pairs with $a \leq b$ and $ab \leq 6$ are $(1,1)$, $(1,2)$, $(1,3)$, $(1,4)$, $(1,5)$, $(1,6)$, $(2,2)$ and $(2,3)$. For $(1,1)$ the equation reads $-c = 4$, and for $(1,2)$ it reads $0 = 6$; both are impossible. For $(1,3)$ we get $c = 8$, for $(1,4)$ we get $2c = 10$ so $c = 5$, and for $(2,2)$ we get $2c = 8$ so $c = 4$. For $(1,5)$ we get $3c = 12$, so $c = 4$, which breaks $b \leq c$; it is $(1,4,5)$ again in another order. For $(1,6)$ we get $4c = 14$ and for $(2,3)$ we get $4c = 10$, neither of which has an integer solution.
So the solutions are the permutations of $(1,3,8)$, $(1,4,5)$ and $(2,2,4)$, and each works: $24 = 2 \times 12$, $20 = 2 \times 10$ and $16 = 2 \times 8$. In the write-up, justify the ordering by symmetry, show the bound and list every case with its outcome, so that a marker can tick each one without doing any arithmetic.

1.5 Writing up "find all"

Two directions.
To answer "find all x with property P" with a set S, you must prove both of the following.

1. Every element of S has property P. This is the checking direction, done by substitution or by construction.
2. Every x with property P lies in S. This is the necessity direction, usually the hard part.

Missing the first direction costs marks. Missing the second means the problem is not solved. The same applies to "for which n is it possible?": you need a construction for the good n and an impossibility proof for the rest.

The checking direction is not a formality. Squaring, multiplying or combining equations can let in false solutions; the 2025 report notes candidates whose derived equations admitted sign combinations that failed the original system.

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Example 3.
For which positive integers n can $\{1, 2, \ldots, n\}$ be split into two sets with equal sums?

Solution.
Necessity is a parity argument. The total is $\frac{n(n+1)}{2}$, and it must be even, so $4 \mid n(n+1)$. Since n and $n+1$ are coprime and only one of them is even, this means $4 \mid n$ or $4 \mid n+1$, that is, $n \equiv 0$ or $3 \pmod 4$.
Sufficiency needs a construction. The block $\{k, k+1, k+2, k+3\}$ splits evenly as $\{k, k+3\}$ and $\{k+1, k+2\}$. If $n \equiv 0 \pmod 4$, cut $1, \ldots, n$ into such blocks and split each. If $n \equiv 3 \pmod 4$, put $\{1, 2\}$ against $\{3\}$ and cut $4, \ldots, n$ into blocks.
The write-up should state the answer, then prove necessity and sufficiency under separate headings. A script with only one of the two is marked "0 plus".

1.6 An annotated model solution

Example 4.
Find all pairs (m, n) of positive integers such that $m+n$ divides both m^2+n and n^2+m.

Solution.
In rough: $(1,1)$ and $(3,3)$ work, while $(2,2)$ and $(1,2)$ fail. A guess: $m = n$ with m odd. Note that modulo $m+n$ we can replace n by $-m$. I have annotated the write-up below with remarks on how a marker reads it.
Answer. The pairs (m, m) with m odd.
Marker's eye: the answer comes first, so the marker knows what is being proved.
These pairs work. If $m = n$ is odd, then $m+n = 2m$ and $m^2+n = m(m+1)$. As $m+1$ is even, $m(m+1) = 2m \cdot \frac{m+1}{2}$ is a multiple of $2m$. The second condition is identical, since $m = n$.
Marker's eye: the checking direction is short, so do it first and secure it.
No other pairs work. Let $s = m+n$ and suppose s divides m^2+n and n^2+m. Since $n \equiv -m \pmod s$,

$$m^2+n \equiv m^2-m = m(m-1) \quad \text{and} \quad n^2+m \equiv m^2+m = m(m+1) \pmod s.$$

So s divides both $m(m-1)$ and $m(m+1)$, and therefore their difference, $2m$.
Now $2m > 0$, and $2m < 2s$ because $n \geq 1$. The only multiple of s that is greater than 0 and less than $2s$ is s itself, so $2m = m+n$, giving $n = m$. Finally, with $n = m$ the condition is $2m \mid m(m+1)$, and dividing by m this says $2 \mid m+1$, so m is odd.
Marker's eye: this paragraph is the hard step. Writing "$s \mid 2m$, so $s = 2m$" with no reason would be a gap, because a divisor of $2m$ can be smaller than $2m$. The bound $0 < 2m < 2s$ is what justifies the step, and it is written out.
Conclusion. The pairs are exactly (m, m) with m odd.
Marker's eye: the conclusion repeats the answer in the form the question asked for.

Common ways to lose marks.
(all drawn from recent BMO1 markers' reports):

- giving the answer with no proof that nothing else works, or never checking that the claimed answers work;

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- dividing by a quantity that might be zero;
- missing cases, often small ones such as $p = 2$ or $n = 1$;
- assuming a special case, such as treating real variables as integers;
- completing a table by pattern and calling it a proof;
- claims like "exponentials grow faster" with no starting point;
- geometry with angle facts that are stated but not justified by a named theorem;
- misreading the question, and so proving something else.

1.7 How to spot this in a contest

On BMO the write-up is what is marked. Before you write, know the answer and the outline of the proof, and leave real time for writing: a solution scrawled out in the last five minutes is often marked "0 plus".

When you finish, reread as a marker and check the following.

1. The answer is stated.
2. Both directions are proved.
3. Every division is safe.
4. The cases are exhaustive, with small ones shown.
5. Every "clearly" is justified.

Fixing a gap now costs two minutes; leaving it can cost five marks.

2 Practice problems

The problems are roughly in order of difficulty, from ⋆ to ⋆⋆⋆⋆. Hints are in Section 4 and full solutions in Section 5.

Problem 1 ⋆ Find all positive integers n for which $n^4 + 4$ is prime.

A student writes: *"We have* $n^4 + 4 = (n^2 + 2)^2 - (2n)^2 = (n^2 + 2n + 2)(n^2 - 2n + 2)$. *Both factors are greater than* 1, *so* $n^4 + 4$ *is never prime, and there are no such* n."

Find the gap, correct the answer and write a complete solution.

Problem 2 ⋆⋆ Find all functions $f : \mathbb{Z} \to \mathbb{Z}$ such that

$$f(x + y) = f(x) + f(y) + 2xy$$

for all integers x and y. Write a complete solution to BMO standard.

Problem 3 ⋆⋆ Find all real solutions (x, y) of the system

$$x^2 y = 4y, \qquad x + y^2 = 7.$$

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A student writes: *"Dividing the first equation by y gives $x^2 = 4$, so $x = 2$ or $x = -2$. If $x = 2$ then $y^2 = 5$, and if $x = -2$ then $y^2 = 9$. The solutions are $(2, \sqrt{5})$, $(2, -\sqrt{5})$, $(-2, 3)$ and $(-2, -3)$."*
Find the gap and give a complete solution.

Problem 4 ⋆⋆⋆ Prove that for every positive integer n,

$$1 + \frac{1}{2^2} + \frac{1}{3^2} + \cdots + \frac{1}{n^2} < 2.$$

A student writes: *"Let S_n be the sum. Then $S_1 = 1 < 2$. Suppose $S_k < 2$. Then $S_{k+1} = S_k + \frac{1}{(k+1)^2} < 2 + \frac{1}{(k+1)^2}$, and since $\frac{1}{(k+1)^2}$ is very small, $S_{k+1} < 2$. By induction, $S_n < 2$ for all n."*
Explain why this is not a proof, and write a correct one.

Problem 5 ⋆⋆⋆ Determine all positive integers n for which the set $\{1, 2, \ldots, n\}$ can be split into three sets with equal sums. Write a complete solution to BMO standard.

Problem 6 ⋆⋆⋆⋆ Find all pairs (a, b) of positive integers such that $a^2 + b$ and $b^2 + a$ are both powers of 3. Write a complete solution to BMO standard.

3 Contest ladder

Real past problems on the ideas of this unit, easiest first. Each source is a link to the official paper or the AoPS wiki; the note under it says where to start, not how to finish.

1. **UKMT BMO1 2022 Problem 1** ⋆⋆
   Find all: the hard part is proving your list is complete
2. **UKMT BMO1 2023 Problem 4** ⋆⋆
   Factorise, then handle coprime factors with care
3. **USAJMO 2011 Problem 1** ⋆⋆
   Different arguments for different parities of n; write both
4. **UKMT BMO1 2025 Problem 2** ⋆⋆⋆
   Beware dividing by a difference, and check candidates in the original system
5. **UKMT BMO1 2025 Problem 1** ⋆⋆⋆
   Constructions for most cases, impossibility proofs for the rest
6. **USAJMO 2023 Problem 1** ⋆⋆⋆
   Simplify, factorise, then a careful case check of divisors
7. **UKMT BMO1 2024 Problem 2** ⋆⋆⋆
   Prove the trick works for some n and fails for every other n
8. **USAJMO 2013 Problem 1** ⋆⋆⋆
   Work modulo 9 and keep the case analysis tidy
9. **UKMT BMO1 2023 Problem 2** ⋆⋆⋆⋆
   Step backwards one term at a time; the induction must be written carefully
10. **USAJMO 2015 Problem 4** ⋆⋆⋆⋆
    Find all functions: derive the form, then verify it

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4 Hints

1. The factorisation is right. Test the claim about the factors on the smallest possible n.
2. Compare with x^2. What equation does $g(x) = f(x) - x^2$ satisfy? Then use induction carefully, not forgetting negative integers.
3. What did the student assume when they divided? Rewrite the first equation as a product equal to zero.
4. The statement $S_k < 2$ is too weak to carry the induction. Try to prove something stronger, of the form $S_n \leq 2 - (\text{something positive})$.
5. There are two necessary conditions: one from divisibility of the total, and one from the set that contains n. For sufficiency, find a block of six consecutive integers that splits into three pairs with equal sums.
6. Deal with $a = b$ first, then assume $a > b$. One of the two powers of 3 then divides the other, so reduce $a^2 + b$ modulo $b^2 + a$ and count the factors of 3 in what you get.

5 Solutions

Problem 1

Find all positive integers n for which $n^4 + 4$ is prime.

A student writes: "We have $n^4 + 4 = (n^2 + 2)^2 - (2n)^2 = (n^2 + 2n + 2)(n^2 - 2n + 2)$. Both factors are greater than 1, so $n^4 + 4$ is never prime, and there are no such n."

Find the gap, correct the answer and write a complete solution.

The claim that both factors exceed 1 is asserted without proof, and it is false for $n = 1$, where $n^2 - 2n + 2 = 1$. In that case $n^4 + 4 = 5$, which is prime, so the student's answer is wrong.

Here is a complete solution. The answer is $n = 1$. For $n = 1$ we get 5, which is prime. Now let $n \geq 2$. We have $n^4 + 4 = (n^2 + 2n + 2)(n^2 - 2n + 2)$, and $n^2 - 2n + 2 = (n - 1)^2 + 1 \geq 2$ since $n - 1 \geq 1$. The other factor is larger, since it exceeds the first by $4n$. So $n^4 + 4$ is a product of two integers that are each at least 2, and it is not prime. Hence $n = 1$ is the only solution.

Answer: $n = 1$

Problem 2

Find all functions $f : \mathbb{Z} \to \mathbb{Z}$ such that

$$f(x + y) = f(x) + f(y) + 2xy$$

for all integers x and y. Write a complete solution to BMO standard.

Answer. The functions $f(x) = x^2 + cx$, where c is any integer.

These functions work. If $f(x) = x^2 + cx$ then f maps integers to integers, and

$$f(x) + f(y) + 2xy = x^2 + cx + y^2 + cy + 2xy = (x + y)^2 + c(x + y) = f(x + y).$$

No other functions work. Suppose f satisfies the equation and let $g(x) = f(x) - x^2$, which is again a function from $\mathbb{Z}$ to $\mathbb{Z}$. Substituting $f(x) = g(x) + x^2$ and cancelling $(x + y)^2 = x^2 + y^2 +$

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$2xy$ gives $g(x+y) = g(x) + g(y)$ for all integers x, y. Putting $x = y = 0$ gives $g(0) = 2g(0)$, so $g(0) = 0$. Let $c = g(1)$, an integer.

We prove $g(n) = cn$ for all $n \geq 0$ by induction. It holds for $n = 0$. If $g(n) = cn$ then $g(n+1) = g(n) + g(1) = cn + c = c(n+1)$. For negative integers, put $x = n$ and $y = -n$ with $n > 0$: then $0 = g(0) = g(n) + g(-n)$, so $g(-n) = -cn = c \cdot (-n)$. Hence $g(x) = cx$ for every integer x, and $f(x) = x^2 + cx$.

Conclusion. The solutions are exactly $f(x) = x^2 + cx$ with $c \in \mathbb{Z}$.

Problem 3

Find all real solutions (x, y) of the system

$$x^2 y = 4y, \qquad x + y^2 = 7.$$

A student writes: "Dividing the first equation by y gives $x^2 = 4$, so $x = 2$ or $x = -2$. If $x = 2$ then $y^2 = 5$, and if $x = -2$ then $y^2 = 9$. The solutions are $(2, \sqrt{5})$, $(2, -\sqrt{5})$, $(-2, 3)$ and $(-2, -3)$."

Find the gap and give a complete solution.

Dividing by y assumes $y \neq 0$, and the case $y = 0$ was never considered. That case contains one more solution.

Here is a complete solution. The first equation is equivalent to $y(x^2 - 4) = 0$, that is, to $y(x - 2)(x + 2) = 0$. A product is zero if and only if one of its factors is, so every solution has $y = 0$, $x = 2$ or $x = -2$, and we find the solutions of $x + y^2 = 7$ in each case.

Case 1: $y = 0$**.** Then $x = 7$, giving $(7, 0)$.

Case 2: $x = 2$**.** Then $y^2 = 5$, so $y = \pm\sqrt{5}$, giving $(2, \sqrt{5})$ and $(2, -\sqrt{5})$.

Case 3: $x = -2$**.** Then $y^2 = 9$, so $y = \pm 3$, giving $(-2, 3)$ and $(-2, -3)$.

Each of these pairs satisfies both equations, because in every case one factor of $y(x - 2)(x + 2)$ is zero and the second equation was solved directly. No pair lies in two cases, since x cannot be both 2 and -2, and $y = 0$ forces $x = 7$. Hence there are exactly five solutions: $(7, 0)$, $(2, \pm\sqrt{5})$ and $(-2, \pm 3)$.

Answer: $(2, \sqrt{5})$, $(2, -\sqrt{5})$, $(-2, 3)$, $(-2, -3)$, $(7, 0)$

Problem 4

Prove that for every positive integer n,

$$1 + \frac{1}{2^2} + \frac{1}{3^2} + \cdots + \frac{1}{n^2} < 2.$$

A student writes: "Let S_n be the sum. Then $S_1 = 1 < 2$. Suppose $S_k < 2$. Then $S_{k+1} = S_k + \frac{1}{(k+1)^2} < 2 + \frac{1}{(k+1)^2}$, and since $\frac{1}{(k+1)^2}$ is very small, $S_{k+1} < 2$. By induction, $S_n < 2$ for all n."

Explain why this is not a proof, and write a correct one.

From $S_k < 2$ the student correctly deduces $S_{k+1} < 2 + \frac{1}{(k+1)^2}$, but that does not give $S_{k+1} < 2$. However small the added term is, it is positive, and nothing in the hypothesis leaves room for it. The phrase "very small" hides the fact that the induction does not close. Instead we prove a stronger statement (strengthening the induction hypothesis).

Claim. For every positive integer n, $S_n \leq 2 - \frac{1}{n}$.

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Proof. For $n = 1$ both sides equal 1. Suppose $S_k \le 2 - \frac{1}{k}$ for some $k \ge 1$. Then

$$S_{k+1} \le 2 - \frac{1}{k} + \frac{1}{(k+1)^2}.$$

We want this to be at most $2 - \frac{1}{k+1}$, which is equivalent to

$$\frac{1}{(k+1)^2} \le \frac{1}{k} - \frac{1}{k+1} = \frac{1}{k(k+1)}.$$

This holds because $k(k+1) < (k+1)^2$. So $S_{k+1} \le 2 - \frac{1}{k+1}$, completing the induction.

Hence $S_n \le 2 - \frac{1}{n} < 2$ for all n. (Equivalently, one can bound $\frac{1}{j^2} < \frac{1}{j-1} - \frac{1}{j}$ for $j \ge 2$ and telescope; the strengthened hypothesis is exactly the telescoped sum.)

Problem 5

Determine all positive integers n for which the set $\{1, 2, \ldots, n\}$ can be split into three sets with equal sums. Write a complete solution to BMO standard.

Answer. Exactly the integers $n \ge 5$ with $n \equiv 0$ or $2 \pmod 3$.

Necessity. The total is $\frac{n(n+1)}{2}$, so each set must have sum $\frac{n(n+1)}{6}$, and this must be an integer. Since $n(n+1)$ is always even, that means $3 \mid n(n+1)$. Since 3 is prime, $3 \mid n$ or $3 \mid n+1$, that is, $n \equiv 0$ or $2 \pmod 3$. Also, the set containing n has sum at least n, so $\frac{n(n+1)}{6} \ge n$, which gives $n \ge 5$.

Sufficiency. We give splittings for $n = 5, 6, 8, 9$:

$n = 5$: $\{5\}$, $\{1,4\}$, $\{2,3\}$, each with sum 5. $n = 6$: $\{1,6\}$, $\{2,5\}$, $\{3,4\}$, each with sum 7. $n = 8$: $\{4,8\}$, $\{5,7\}$, $\{1,2,3,6\}$, each with sum 12. $n = 9$: $\{6,9\}$, $\{7,8\}$, $\{1,2,3,4,5\}$, each with sum 15.

Now suppose $\{1, \ldots, n\}$ has a splitting. The six numbers $n+1, \ldots, n+6$ form the pairs $\{n+1, n+6\}$, $\{n+2, n+5\}$ and $\{n+3, n+4\}$, each with sum $2n+7$. Adding one pair to each set gives a splitting of $\{1, \ldots, n+6\}$.

The integers $n \ge 5$ with $n \equiv 0$ or $2 \pmod 3$ are those with $n \ge 5$ and $n \equiv 0, 2, 3$ or $5 \pmod 6$. In these four residue classes the smallest such n are 6, 8, 9 and 5 respectively, and every other such n is obtained from one of these by adding 6 repeatedly. So every such n has a splitting.

Conclusion. The set can be split into three sets with equal sums exactly when $n \ge 5$ and $n \equiv 0$ or $2 \pmod 3$.

Problem 6

Find all pairs (a, b) of positive integers such that $a^2 + b$ and $b^2 + a$ are both powers of 3. Write a complete solution to BMO standard.

Answer. $(2, 5)$ and $(5, 2)$.

These pairs work. For $(5, 2)$: $5^2 + 2 = 27 = 3^3$ and $2^2 + 5 = 9 = 3^2$. The conditions are symmetric in a and b, so $(2, 5)$ works too.

No other pairs work. Write $a^2 + b = 3^n$ and $b^2 + a = 3^m$. Both numbers are at least 2, so $m \ge 1$ and $n \ge 1$.

If $a = b$, then $a^2 + a = a(a+1)$ is even, so it is not a power of 3. Hence $a \ne b$, and by symmetry we may assume $a > b$ and restore the swapped pairs at the end. Then

$$(a^2 + b) - (b^2 + a) = (a - b)(a + b - 1) > 0,$$

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so $3^n > 3^m$, that is, $n > m$, and therefore 3^m divides 3^n.

Since $a \equiv -b^2 \pmod{3^m}$, we get

$$0 \equiv a^2 + b \equiv b^4 + b = b(b+1)(b^2 - b + 1) \pmod{3^m}.$$

We now count the factors of 3 in $b(b+1)(b^2-b+1)$, according to b modulo 3.

Case $b \equiv 0 \pmod 3$**.** Then $b+1 \equiv 1$ and $b^2 - b + 1 \equiv 1 \pmod 3$, so 3^m divides b. As $b > 0$ this gives $3^m \le b < b^2 + a = 3^m$, a contradiction.

Case $b \equiv 1 \pmod 3$**.** Then $b(b+1)(b^2-b+1) \equiv 1 \cdot 2 \cdot 1 = 2 \pmod 3$, so it is not divisible by 3, contradicting $m \ge 1$.

Case $b \equiv 2 \pmod 3$**.** Now $3 \nmid b$ and $3 \mid b+1$. Write $b^2 - b + 1 = (b+1)^2 - 3b$. Here $9 \mid (b+1)^2$, and $3b \equiv 6 \pmod 9$ because $b \equiv 2 \pmod 3$. So $b^2 - b + 1 \equiv 3 \pmod 9$, which means that $b^2 - b + 1$ contains exactly one factor of 3. Hence the number of factors of 3 in $b(b+1)(b^2-b+1)$ is one more than in $b+1$, and 3^m dividing this product gives $3^{m-1} \mid b+1$. In particular $3^m \le 3(b+1)$.

On the other hand $a \ge b+1$, so $3^m = b^2 + a \ge b^2 + b + 1$. Combining, $b^2 + b + 1 \le 3b + 3$, that is, $(b-1)^2 \le 3$, so $b \le 2$. Together with $b \equiv 2 \pmod 3$ this gives $b = 2$.

Finishing. With $b = 2$ we have $a + 4 = 3^m$ and $a^2 + 2 = 3^n$. Since $a \equiv -4 \pmod{3^m}$, we get $a^2 + 2 \equiv 18 \pmod{3^m}$, and $3^m \mid 3^n = a^2 + 2$, so $3^m \mid 18$ and $m \le 2$. If $m = 1$ then $a = -1$, which is not allowed. If $m = 2$ then $a = 5$, and indeed $a^2 + 2 = 27$ is a power of 3. So $(a, b) = (5, 2)$, which satisfies $a > b$ as assumed.

Conclusion. With $a > b$ the only solution is $(5, 2)$, and $a = b$ is impossible. Restoring the order, the solutions are $(2, 5)$ and $(5, 2)$.

The online version, with progress tracking, is at kevinvariant08.github.io/pathways.html.

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